

Random Vectors and limit theorems

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Motivation

- Example: Two people A and B both flip coin twice.
 X : number of heads obtained by A. Y : number of heads obtained by B. Find $P(X > Y)$.

Joint distributions and marginal distributions

- Discrete case:

Joint probability mass function: $p(x, y) = P(X = x, Y = y)$.

- *Marginal probability mass function* of X can be obtained from the joint probability mass function, $p(x, y)$:

$$p_X(x) = \sum_{y:p(x,y)>0} p(x, y) .$$

Similarly:

$$p_Y(y) = \sum_{x:p(x,y)>0} p(x, y) .$$

Joint distributions and marginal distributions

- Continuous case:

Joint probability density function $f(x, y)$:

$$P\{(X, Y) \in R\} = \int \int_R f(x, y) dx dy$$

- Marginal pdf:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Joint cumulative distribution

- Joint cumulative probability distribution function of X and Y

$$F(a, b) = P\{X \leq a, Y \leq b\} \quad -\infty < a, b < \infty$$

*Conditional distributions

- Discrete case

$$p_{Y|X}(y | x) \triangleq P(Y = y | X = x) = \frac{P(\{X = x\} \cap \{Y = y\})}{P(X = x)}$$

*Conditional distributions

- Continuous case

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Functions of random vectors

- $Z=g(X,Y)$
- $F(z)=P(Z\leq z)= P(g(X,Y)\leq z)$
- Cumulative distribution function methods

Expectation of $g(X, Y)$

- Expectation $E[g(X, Y)]$:

$$= \sum_y \sum_x g(x, y)p(x, y) \quad \text{in the discrete case}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f(x, y)dxdy \quad \text{in the continuous case}$$

- Based on joint distribution, we can derive

$$E[aX + bY] = aE[X] + bE[Y]$$

Extension:

$$\begin{aligned} & E[a_1X_1 + a_2X_2 + \cdots + a_nX_n] \\ &= a_1E[X_1] + a_2E[X_2] + \cdots + a_nE[X_n] \end{aligned}$$

Expectation of $g(X, Y)$

- Example: $E[X]$, X is binomial with n, p :

$$X_i = \begin{cases} 1 & \text{ith flip is head} \\ 0 & \text{ith flip is tail} \end{cases}$$

$$X = \sum_{i=1}^n X_i, E[X] = \sum_{i=1}^n E[X_i] = np$$

Independent random variables

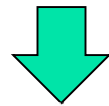
- X and Y are *independent* if

$$P(X \leq a, Y \leq b) = P(X \leq a)P(Y \leq b)$$

- Equivalently: $F(a, b) = F_X(a)F_Y(b)$.
- Discrete: $p(x, y) = p_X(x)p_Y(y)$.
- Continuous: $f(x, y) = f_X(x)f_Y(y)$.
- Proposition : If X and Y are independent, then for function h and g , $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.

Independent and identically
distributed random variables
(i.i.d)

$$(X_1, X_2, \dots, X_n)$$



sample statistic: $g(X_1, X_2, \dots, X_n)$

Covariance

- Definition: *Covariance* of X and Y

$$Cov(X, Y) = E[(X - E(X))(Y - E(Y))]$$

- $Cov(X, X) = E[(X - E(X))^2] = Var(X)$.
- $Cov(X, Y) = E[XY] - E[X]E[Y]$.
- If X and Y are independent, $Cov(X, Y) = 0$.

- Properties:

1. $Cov(X, X) = Var(X)$
2. $Cov(X, Y) = Cov(Y, X)$
3. $Cov(cX, Y) = cCov(X, Y)$
4. $Cov(X, Y + Z) = Cov(X, Y) + Cov(X, Z)$

Correlation

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Sum of independent random variables

- If X_i 's are independent, $i = 1, 2, \dots, n$

$$Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i)$$

$$Var\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 Var(X_i)$$

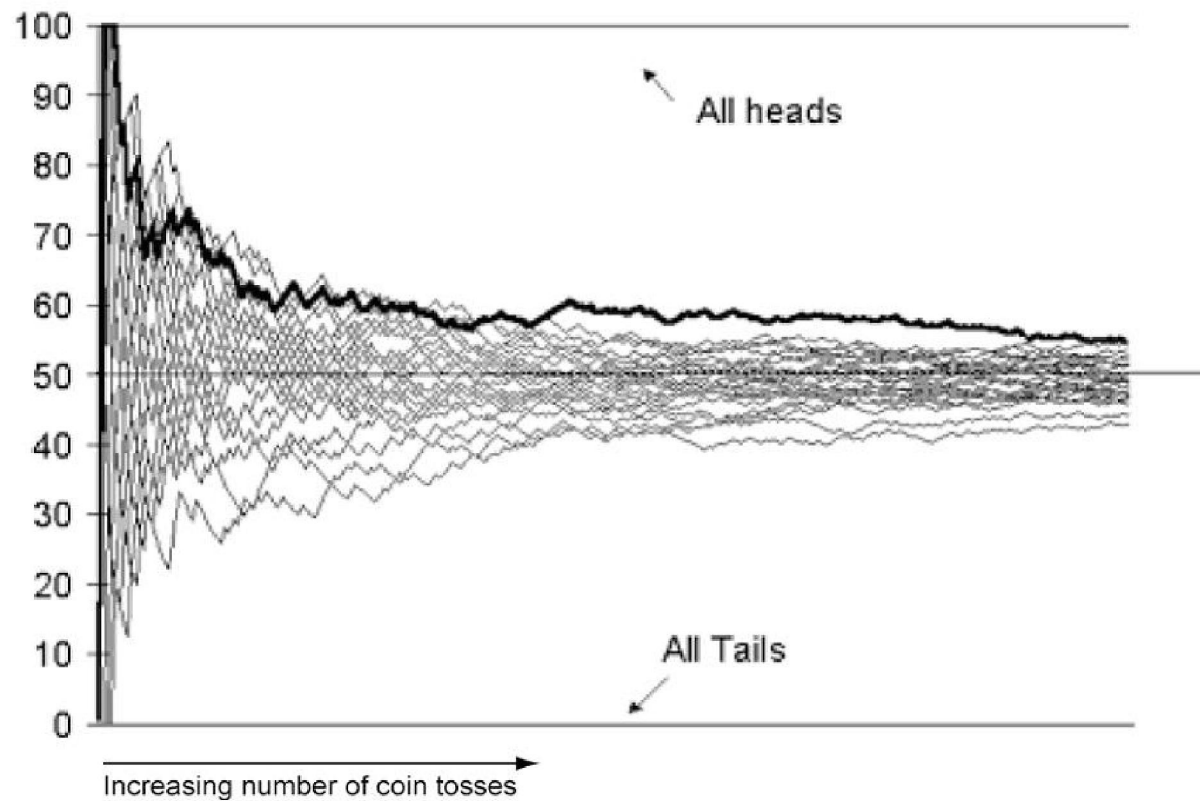
Limit of sum of i.i.d random variables

- Law of Large Numbers
- Central Limit Theorem

Law of large numbers

- “Law of large numbers” is expressing the idea that as the number of trials of a random process increases, the percentage difference between the expected and actual values goes to zero.

Law of large numbers



Weak Law of Large Numbers (WLLN)

- Khinchin's law

Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of i.i.d. r.v.s with mean $E(X_n) = \mu \in (-\infty, \infty)$. Then,

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu.$$

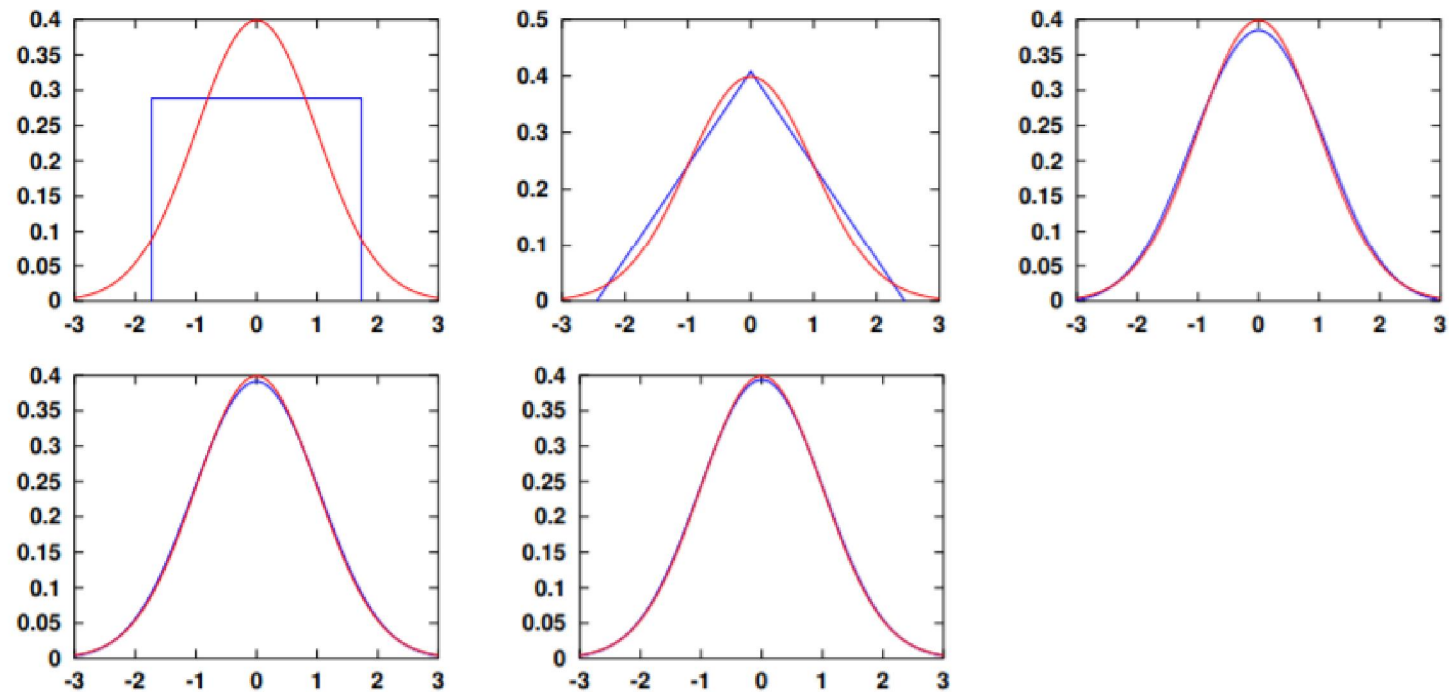
Central limit theorems

$$S_n = X_1 + X_2 + \dots + X_n = \sum_{i=1}^n X_i$$
$$\overline{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{S_n}{n}.$$

Central Limit Theorem: For large n ,

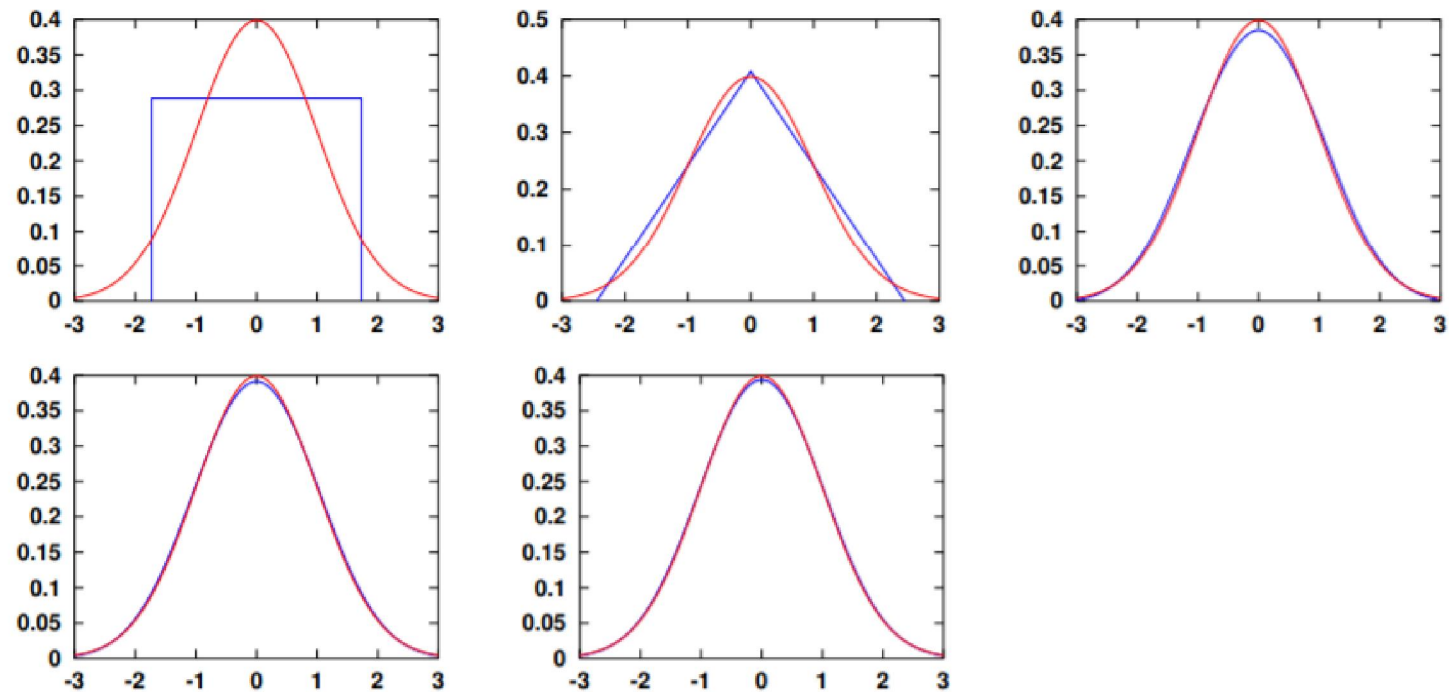
$$\overline{X}_n \approx N(\mu, \sigma^2/n), \quad S_n \approx N(n\mu, n\sigma^2)$$

Central limit theorems



e.g. Uniform distribution

Central limit theorems



e.g. Uniform distribution

Central limit theorems

The Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with expected value $EX_i = \mu < \infty$ and variance $0 < \text{Var}(X_i) = \sigma^2 < \infty$. Then, the random variable

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sqrt{n}\sigma}$$

converges in distribution to the standard normal random variable as n goes to infinity, that is

$$\lim_{n \rightarrow \infty} P(Z_n \leq x) = \Phi(x), \quad \text{for all } x \in \mathbb{R},$$

where $\Phi(x)$ is the standard normal CDF.

Central limit theorems

Assumptions:

- $X_1, X_2 \dots$ are iid Bernoulli(p).
- $Z_n = \frac{X_1 + X_2 + \dots + X_n - np}{\sqrt{np(1-p)}}$.

We choose $p = \frac{1}{3}$.

$$Z_1 = \frac{X_1 - p}{\sqrt{p(1-p)}}$$

PMF of Z_1



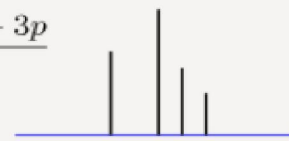
$$Z_2 = \frac{X_1 + X_2 - 2p}{\sqrt{2p(1-p)}}$$

PMF of Z_2



$$Z_3 = \frac{X_1 + X_2 + X_3 - 3p}{\sqrt{3p(1-p)}}$$

PMF of Z_3



$$Z_{30} = \frac{\sum_{i=1}^{30} X_i - 30p}{\sqrt{30p(1-p)}}$$

PMF of Z_{30}

